

Human-Computer Interaction

Project

The project is an individual assignment completed during the semester through in-class project sessions and independent work. Its objective is to design and implement a small HCI-related system, interface, or interaction concept. Students are responsible for selecting their own topic, obtaining approval from the lecturer, implementing the proposed solution, preparing the required documentation, and presenting the final results.

1. Topic Selection and Approval

- Propose an HCI-related project topic of your choice (you may draw inspiration from project examples from previous years)
- Present your idea on the class forum and briefly justify its relevance to HCI.
- Wait for lecturer's approval before proceeding. You may be asked to refine or adjust your concept.

2. Initial Project Description (max. 1 page)

After approval, submit a one-page project description that includes:

- **Project objective:** What problem you address and the intended user interaction.
- **Input/Output specification:** What data the system receives and what it produces.
- **Planned functionality:** Key features and intended user experience.
- **Tools and environment:** Programming languages, libraries, frameworks, and hardware (if applicable).
- **Constraints and assumptions:** Technical limitations, performance constraints, interaction boundaries, etc.

3. Project Implementation

You will develop your system during the project labs and through independent work. Your implementation should follow the approved design but may evolve as you refine your approach. Ensure your solution is functional and demonstrably interactive.

4. Final Documentation (4–5 pages)

Submit a concise implementation report expanding the initial description. It should include:

- **Updated project assumptions** with any modifications.
- **System architecture / processing pipeline:** A clear description or diagram of data flow.
- **Main algorithm(s) and technical approach:** Explanation of key methods.
- **User interface design:** Screenshots or illustrations of the GUI or interaction flow.
- **Execution instructions:** How to run the system, including software/hardware requirements.
- **Example results:** Demonstrations, test cases, screenshots, or performance observations.

5. Demonstration Video

Prepare a short video demonstrating your system in action. The video should:

- Clearly show the core functionality and user interaction flow.
- Highlight the most important features or technical elements.

6. Final Presentation / Defense

At the end of the course, you will present your project to the group. During the presentation, you should discuss the project assumptions, the overall concept, and the intended interaction. You must present the demonstration video or provide a live demonstration of your system. Additionally, you are expected to explain the key technical decisions, describe the main implementation challenges, and conclude with a brief reflection on limitations and potential future improvements.

The project documentation, source files, and demonstration video must be submitted by e-mail to: anowosielski@zut.edu.pl

In the e-mail title, please use the scheme:

- HCI Your Full Name